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Breaking the Stigma of Epilepsy

A story that is more than just a seizure

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Abstract

When will the next seizure occur? How will it happen? Fear, anxiety, depression. Three factors that surround epilepsy patients every day. Not knowing when the next seizure will occur, how people around them will react, and the fear of losing their lives are challenges that people living with epilepsy face every day. Numbers are increasing as $\frac{1}{3}$ of epilepsy cases go untreated for up to 3 years after diagnosis (Push10 Webmaster) and 70% of patients could live seizure-free if properly treated and diagnosed (World Health Organization). Through research-based explanations, this article explores the causes, symptoms, diagnosis, treatments, and lifestyle guidelines for living with epilepsy. So let's give hope to these patients: the hope for a better future and the understanding that epilepsy doesn't define who they are.

Introduction

"Dear Epilepsy, Just so you know, I won't give up. I won't give in. I have endless hope. Endless faith. You may be in my life. But you don't own my life (Epilepsy Network). According to NIH PubMed, epilepsy is the 4th most common brain disorder and affects around 65 million people worldwide. It is made up of recurring and unprovoked seizures caused by abnormal electrical activity in the brain. There are 3 types that distinguish the severity of the seizure: generalized epilepsy, where both sides of the brain are affected; focal epilepsy, with only one affected area of the brain; and unknown, where the major source of activity is not identified.

For millions of people affected by epilepsy, these seizures happen unexpectedly and sometimes without any warning signs. Untreated epilepsy can result in various life-threatening factors, from social disability to death. Since seizures are unpredictable, friends and family that are surrounding the epilepsy patients are a significant contributor in helping manage their fears. According to NIH PubMed, epilepsy causes a 4-7 times higher mortality rate than others because of the major source of electrical activity. So let's give these patients the chance to explore the world by creating a safe environment for all.

Causes

Firstly, what causes epilepsy and what are the risk factors that increase the chances of being diagnosed with this medical condition? According to Epilepsy Research Institute UK, 65% of people living with epilepsy are not aware of the causes of their diagnosis. The remaining 35% are generally caused by anything that disrupts the normal pattern of nerve cell activity in the brain: genetic influence, developmental conditions, and head trauma.

Genetic influence, whether it is inherited or a change to a gene's DNA sequence, plays a major role in epilepsy diagnosis. For example, a type of epilepsy is the channelopathy-associated epilepsy. This is when the DNA sequence changes the ion channels, which are proteins that control the signals of the neurons in the brain (NIH). When modified, multiple seizures are triggered because the neurons become hyperexcited and fire too frequently.

Developmental conditions, like Attention Deficit Hyperactivity Disorder (also known as ADHD), are shown to increase children's risk of certain types of epilepsy. This includes generalized epilepsy (where the whole brain is affected) and frontal lobe epilepsy

(where there is only one affected part of the brain) (Ahmed et al., 2022). A study from the International League Against Epilepsy shows that ADHD was present in 30-40% of epileptic children. This is because people diagnosed with this disorder may have functional abnormalities in multiple areas of the brain that disrupt the pattern of the nerve cell activity. Other relationships that are common between ADHD and epilepsy patients are inflammation, anxiety, and stress.

Post-traumatic epilepsy (PTE) is the type of epilepsy developed after a traumatic brain injury (TBI), also known as head trauma. It can be an immediate seizure that happens within twenty-four hours of the injury, an early seizure that is less than a week, or a late seizure after a week of the injury (Kan Ding et al., 2016). A seizure occurs because the brain is traumatized, inflamed, and disrupted. For these reasons, there are multiple factors that trigger epilepsy, from genetics to daily lifestyles.

Signs & Symptoms

So what are the warning signs of a seizure or epilepsy? How can the people surrounding you as well as yourself be prepared? Since each person's brain functions differently, the body reacts in various ways. Therefore, the signs, symptoms, and the type of seizure all depend on the person (Mayo Clinic, 2025).

General symptoms that might occur during an epileptic seizure include a short-term confusion, stiff muscles, and emotional distress like fear, anxiety, or déjà vu (Mayo Clinic, 2025). Emotional distress mostly happens to patients with focal epilepsy, which, as mentioned above, affects one part of the brain: the temporal lobe. This lobe is responsible for short-term memory and processing emotions (Mayo Clinic, 2025). Thus, with the temporal lobe affected, it causes the patient's body to react with these symptoms.

The signs that are visible to the people surrounding the patient are a brief moment when the person doesn't move or respond, jerking movements of arms and legs, and a loss of consciousness (Mayo Clinic, 2025). During an epileptic seizure, patients might lose consciousness as the areas of the brain responsible for thinking and awareness are malfunctioning (Blumenfeld & Taylor, 2003). Some patients might experience jerking movements because of the muscles that contract repeatedly (NIH). This is because there is an excessive firing of electrical signals in the brain that overwhelm its communication with the body's muscles.

Note: Symptoms vary widely between individuals; readers should not self-diagnose based on this list. If someone suspects they have this condition, they should consult a healthcare provider.

Diagnosis

Epilepsy or seizure? With the rapid advancement of technology, the medical field has developed various testing methods to diagnose different patients and improve their quality of life: electroencephalography (EEG), neuroimaging (MRI scans), and single-photon emission computerized tomography (SPECT).

Electroencephalography, also known as EEG, is a testing method that records the brain's electrical activity. During an EEG test, electrodes (electrical conductors) are placed on the patient's scalp, allowing health professionals to monitor the patient's brain activity during sleep or speech. If the patient is diagnosed with epilepsy, the EEG amplitudes will increase compared to an unaffected human being because of the rapid and uncontrollable neuronal discharge (Blinowska, 2006).

Neuroimaging, which includes MRI scans (magnetic resonance imaging) of the brain, is shown to be very efficient in identifying epilepsy. Other imaging methods like CT scans (computerized tomography) show a less detailed view of the patient. According to NIH, MRI scans were able to successfully diagnose 80% of focal epilepsy patients (Roy & Pandit, 2011). Unfortunately, this type of brain scanning is unavailable for a large number of patients because of its elevated cost and the long scanning period.

Single-photon emission computerized tomography, also known as SPECT, is another type of medical imaging of the brain. This method is used if both CT and MRI scans weren't able to identify the origin of the seizure. SPECT is performed through injecting a low dose of radioactive substance into the vein of the patient. The contrast creates a 3D map of the blood flow, which shows which part of the brain receives a higher amount of blood and most likely identifies the source of the seizures. By visualizing the internal functions, it can help doctors further investigate the patient's diagnosis (Mayo Clinic, « Epilepsy - Diagnosis and Treatment - Mayo Clinic », 2025).

Treatment and Management

But what can I do as an epilepsy patient or as a friend or family member of someone going through epilepsy? Living with this condition can be tough, but many factors from

lifestyles to drugs to surgeries have been adapted to enrich the quality of their lives (Perucca et al., 2018).

Antiepileptic drugs, also known as AEDs, are the main approach to epilepsy treatment. According to PubMed NIH, antiepileptic drugs help $\frac{2}{3}$ of newly diagnosed patients reduce seizures. Without disturbing any non-epileptic activity and normal function of the nervous system, these drugs modify the excitement of neurons. This reduces uncontrolled activity in the brain by using sodium and calcium channels, which are proteins that control the electrical activity in the body (Rogawski & Löscher, 2004).

According to Wiley, if two appropriate trials of this drug have failed, epilepsy surgery is the alternative treatment (Perucca et al., 2018). This may be because of the patient having drug-resistant epilepsy, where medication no longer works in treating the disorder. In terms of surgery, there are many types of procedures that can be performed based on the type of epilepsy and the degree of severity: resective surgery, which removes the part of the brain where the seizures originated; and Multiple Subpial Transection, known as MST, where brain tissue is cut to disrupt the spread of seizures (Patient Education Library).

For patients where surgery has failed to treat epilepsy, lifestyle therapies have been developed to control these seizures. For instance, the ketogenic diet is an option recommended mostly for children. This diet includes a heavier intake of glucose and ketones. Glucose is a sugar used to be converted to energy, while ketones are chemicals the body produces to break down fat into energy. Thus, this meal plan is made to promote the production of energy through healthy fat and reduced carbohydrates (Saxena & Nadkarni, 2011).

Note: Treatment plans are highly individualized; readers should always consult their personal physician before starting or modifying treatment.

Living With Epilepsy

What can you do to protect yourself and others around you in case of an epileptic seizure? Or in prevention of one? A healthy lifestyle, following the medication, and avoiding unprescribed medicine all shape the recommendations involved with improving the quality of life when living with epilepsy (Murray, 2017).

First, taking the prescribed amount of medication made by your doctor is important to

assure a constant routine. Next, a healthy lifestyle is important to reduce stress, such as increasing sleep, following a balanced diet, and decreasing harmful use of drugs or alcohol. Additionally, avoiding medications that can possibly increase the severity of the condition can pose greater benefits (Murray, 2017). For example, certain antibiotics may interact harmfully with anti-seizure medications or even antidepressants (found in 30-50% of epilepsy patients according to NIH PubMed).

Among the causes of this condition, there are factors that you should be aware of to prevent epilepsy: traumatic brain injury (TBI) and stroke. Traumatic brain injury is caused by an outside force like a blow to the head during sports injuries or if an object enters the brain tissues (National Institute of Neurological Disorders and Stroke). It is the cause of epilepsy for around 5% of total cases and is most common in children because they tend to have thinner and softer skulls (Thurman et al., 2020). So next time you do a sport (especially combat sports), don't forget to protect yourself.

Stroke occurs when the brain has less access to oxygen and nutrients because of the interrupted blood flow. This factor is most common for older adults in 19-24% of cases because as they age, they tend to have stiffer blood vessels and an overall decline in the heart's health. For all ages, high blood pressure and high cholesterol all increase the chances of epilepsy. Therefore, although half of epilepsy patients have unknown causes, there are factors in your lifestyle that can prevent the diagnosis of this condition and make sure your body stays healthy (National Institute of Aging).

Conclusion

This is the reality for millions of epilepsy patients every day. Epilepsy isn't just a seizure: the anxiety before, the isolation after, and the depression that follows all shape the quality of their lives. So if you or someone you love experience one or more symptoms of epilepsy, don't hesitate to reach out to your doctor. This helps them diagnose the cause of the symptoms to prevent the worsening of this condition.

To learn more about epilepsy, reliable resources include NIH (PubMed), National Institute for Health and Care Excellence (NICE), and Epilepsy Canada. So as a team, let's work together to raise public awareness on this medical condition and the millions of epilepsy patients that face life-threatening challenges every day.

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All claims in this article are supported by evidence-based sources.

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